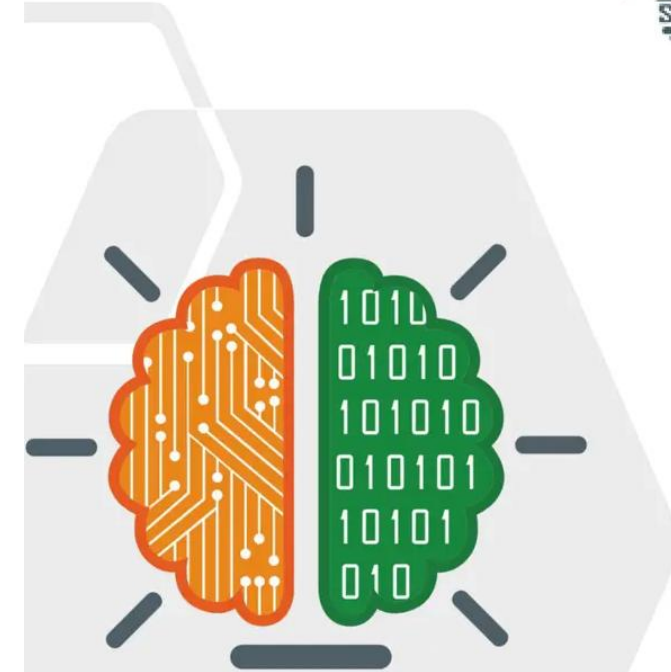


SMART INDIA HACKATHON 2026

RaahSetu

Problem Statement ID –	SIH26002
Problem Statement Title –	AI-Based Smart Logistics & Accessibility Intelligence Platform for North Eastern Region
Theme –	Disaster Management / Smart Automation
PS Category –	Software
Team Name (Registered on portal) –	[Enter your Team Name here]

Idea Title: RaahSetu — Explainable Risk-Aware Logistics Routing for North-East India



RaahSetu: Geospatial Logistics Intelligence Platform



Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation: RaahSetu is an operations dashboard with a custom Risk-Aware A* routing engine. Instead of only finding the shortest path, it avoids verified landslide blackspots, flood zones and poor road surfaces — and shows the evidence behind every route decision.

How it addresses the problem: Standard apps route heavy logistics into danger zones in the North-East because they optimise only for speed. RaahSetu quantifies the time-versus-safety tradeoff, respects vehicle limits, and raises isolation alerts for SDRF/NDRF when the only access road to a district is verified as blocked.

Innovation and uniqueness

- ❖ Not a Google-Maps wrapper: open OSMnx graph networks with directed parallel edges preserved.
- ❖ Risk enters the cost function itself: $c(e) = t(e) \times [1 + 4\lambda \cdot r(e)]$ — admissible A* stays provably optimal for every operator setting.
- ❖ Evidence-first closures: field reports carry corroboration counts; only reviewer-verified closures change routing.
- ❖ Offline-first field reporting (IndexedDB queue + photo evidence) with automatic sync for low-connectivity corridors.

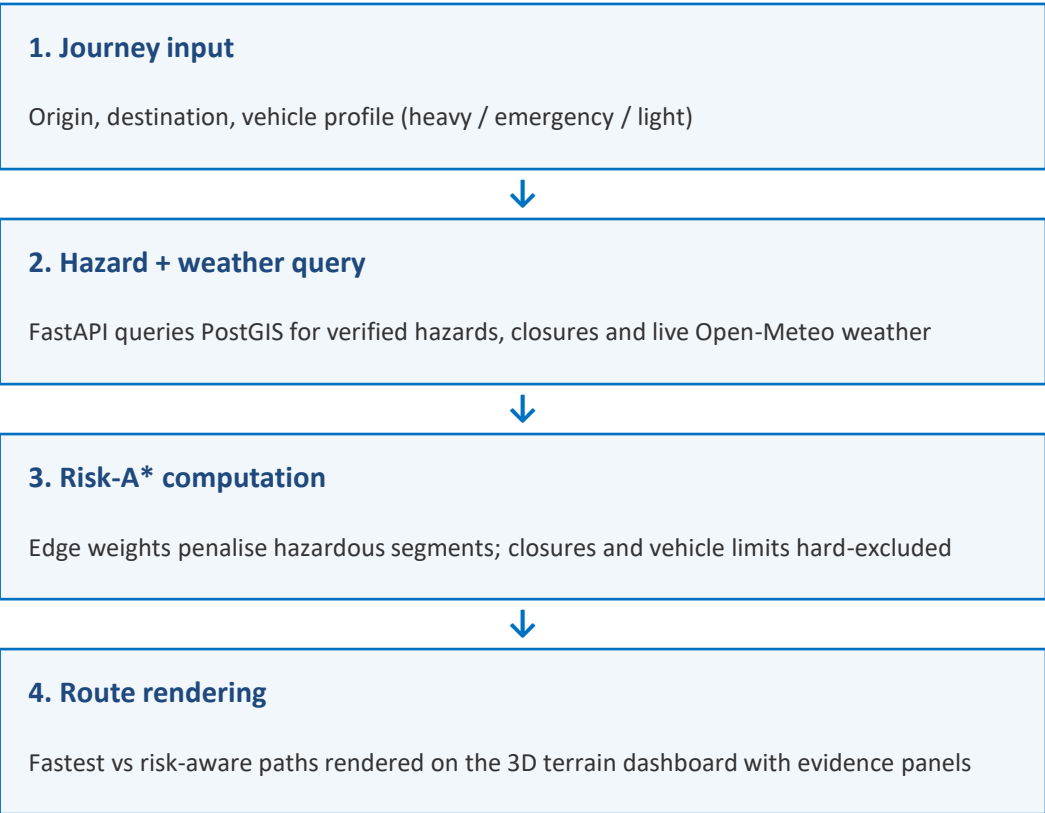
Working today: 8-state OSM data acquired (273,494 road ways), Guwahati corridor pilot on 13,681 directed segments, 45 automated tests, 300 A/Dijkstra oracle checks.*

Technical Approach

Technologies to be used

Frontend	React, TypeScript, React Three Fiber (WebGL 3D terrain), installable PWA
Backend	Python 3.12, FastAPI, OSMnx (OpenStreetMap graphing)
Database	Supabase PostgreSQL + PostGIS spatial engine, row-level security
Deployment	Docker, Kubernetes on cloud VMs, versioned graph storage

Methodology and process



Verified correctness: the custom A* matches an independent Dijkstra baseline on 300 randomized vehicle/weather scenarios; API contract, security and closure workflows covered by 45 automated tests.

Feasibility and Viability



Analysis of feasibility

- ❖ Highly feasible and cost-effective: open-source GIS data (OpenStreetMap) and keyless weather (Open-Meteo) keep operating cost near zero for government scaling.
- ❖ Cloud database usage measured at ~51 MB on the free tier — raw map data stays in versioned files, only hazards, reviews and fleet state enter PostGIS.
- ❖ Live end-to-end workflow already proven on the cloud database: report → review → closure → alert → reroute → delivery tracking (14/14 steps).

Potential challenges and risks

- ❖ Zero or poor network connectivity across remote North-East corridors.
- ❖ Heavy 3D WebGL rendering on outdated control-room computers.
- ❖ Verified coordinate-level blackspot data is not publicly released for all states.

Strategies for overcoming these challenges

- ❖ Installable PWA with service workers and IndexedDB offline queue; reports and photos sync automatically when connectivity returns.
- ❖ prefers-reduced-motion support, lazy-loaded 3D chunks and an SVG map fallback; UI is decoupled from the API so alerts work regardless of rendering.
- ❖ Honest-data architecture: unverified evidence is never auto-applied; eDAR/MoRTH access is the scaling path, with a reviewer workflow already shipped.

Impact and Benefits



Potential impact on the target audience

Empowers disaster-management authorities (NDRF/SDRF), border logistics and commercial fleets with terrain-aware routing intelligence, ensuring relief and essential supplies reach isolated districts safely — in a region where NH-6 alone is blocked 15–20 times each monsoon.

Social	Prevents heavy vehicles from traversing active landslide zones or damaged bridges; isolation alerts trigger relief coordination for cut-off districts.
Economic	Avoids days of stranding and rework on corridors where logistics runs 20–30% costlier than mainland India; pre-computed viable alternatives cut delay and fuel penalties.
Strategic	Historical choke-point analysis supports predictive stockpiling before extreme weather; every closure decision is auditable with evidence and reviewer identity.

Beneficiaries: NDRF / SDRF relief fleets · Health-supply chains · Commercial heavy logistics · State transport authorities · Drivers on unsafe corridors

Research and References

Core architecture validation	“Route Map AI: An Intelligent Route Optimization System Integrating GIS and Machine Learning” (2026) — validates GIS + risk-weighted pathfinding for safer trips under hazard conditions.	Academic study, 2026
Data sources	OpenStreetMap North-East zone extracts (ODbL 1.0); NASA Global Landslide Catalog (467 historical NER records with positional accuracy); Open-Meteo live weather.	openstreetmap.org · nasa.gov · open-meteo.com
Government datasets	Ministry of Road Transport and Highways (MoRTH) road-safety statistics and eDAR accident records — consulted for context; deliberately not mapped to specific roads without verified coordinates.	morth.nic.in
Algorithmic framework	Graph-theory pathfinding in spatial networks: A* with admissible heuristics vs Dijkstra baseline — verified on 300 randomized cases across vehicle and weather settings.	Dijkstra (1959) ; Hart et al. (1968)
Prototype evidence	Working codebase with 45 automated tests, live cloud end-to-end workflow record, and reproducible route-evaluation suite (56 journeys).	Reproducible open-source prototype (repository shared on request)